

Cosmic Intelligence Model: Formal Logical and Symbolic Representation

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Abstract

This document provides the formal first-order logical and symbolic axiomatization of Constitutive Intelligence (CI) as a foundation for the empirical and philosophical framework developed across the companion papers. The philosophical argument is in Van Slyke (2026a). The empirical biological measurements grounding the axioms are presented in Van Slyke (2026b). The formal cybernetic proof is in Van Slyke (2026c). The Anchor-Single pair confirmation is in Van Slyke (2026d). A central addition in this revision is Lemma 0 — a bridging derivation establishing that Axiom 1 (the existence of Int_0) is not a bare assumption but a logical consequence of the empirically confirmed conservation principle and the universal nonzero constraint density $\Delta C > 0$. Lemma 0b further establishes that the mathematical comparator is logically prior to all laws.

Series Context

This document is Paper 5 (Van Slyke, 2026e) in the Constitutive Intelligence series. The philosophical entry point is Van Slyke (2026a). The empirical biological measurements grounding the axioms are presented in Van Slyke (2026b). The formal cybernetic proof is in Van Slyke (2026c). The Anchor-Single pair confirmation is in Van Slyke (2026d). The domain-agnostic methodology framework is in Van Slyke (2026f). The convergence analysis is in Van Slyke (2026g). The geometry-of-absence analysis is in Van Slyke (2026h).

1 Definitions

Primitive Entities

Lemma 0 (Conservation Ground of Axiom 1)

This lemma establishes that the existential assertion of Int_0 in Axiom 1 is not a bare foundational assumption but follows with logical necessity from two independently established premises: the empirically confirmed universal nonzero constraint density and the conservation principle.

Premise L1 - Universal Nonzero Constraint Density (Empirical)

$$\Delta C(N) = 1 - \frac{H_{\text{observed}}(N)}{H_{\text{max}}(N)} > 0$$

for all measured complexity levels N and all measured biological systems.

Symbol	Definition
Int_0	Pre-existing / Constitutive Intelligence (fundamental source)
Int_e	Emergent intelligence (expression of Int_0)
O	Order (structured, law-bound system)
L	Laws (constraints on all events)
S	Source information (all constraints, initial conditions, laws)
E	Event
R	Apparent randomness
C	Constraints fully known
D	Differentiation (discernment of possibilities)
Rk	Recognition (awareness of patterns)
$\mathcal{A}$	Actualizers (systems capable of differentiation and recognition)
$\mathcal{T}$	Temporal/spatial coordinates
$proj$	Projection operator
θ	Manifestation threshold
ΔC	Constraint density: $1 - H_{\text{observed}}/H_{\text{max}}$ (empirical, Van Slyke 2026b)
κ	Mathematical comparator

Confirmed across 16 independent measurements: four organisms (E. coli K-12, Human BRCA1, S. cerevisiae Chr1, Human COX1 mitochondrial) at four complexity levels (mononucleotide, dinucleotide, codon, amino acid). $\Phi_{CI} = \sup\{\Delta C(N)\} = 0.2080$ (Human BRCA1, codon level). No system retreats to the maximum entropy baseline. Structure is present at every measurable level.

Empirical source: Van Slyke (2026b), Section 3.2. Anchor-Single pair confirmation at 61,320,087 observations: Van Slyke (2026d).

Premise L2 - Conservation Principle (Logical / Thermodynamic)

$$\forall x \forall y (\text{Produces}(x, y) \rightarrow I(y) \leq I(x))$$

No system produces output whose information content exceeds its source. This is simultaneously a thermodynamic necessity (First Law: energy is conserved), a logical necessity (a system cannot output what it does not contain), and a mathematical necessity (Shannon entropy cannot increase without external input). It has never been successfully challenged in physics, mathematics, or logic.

Premise L3 - Intelligence Advances in Matter (Observed)

$\Delta C > 0$ at inception $\wedge \Delta C$ increases with organisational complexity across biological history.

The measured progression from $\Delta C = 0.0592$ at mononucleotide level to $\Delta C = 0.2080$ at the codon level in BRCA1 demonstrates that organisational intelligence accumulates in matter. This is not contested.

Derivation

1. By L1: $\Delta C(N) > 0$ at inception. Structure was present before any accumulation of complexity. It cannot have retreated to zero and reappeared, since L2 forbids the generation of constraint from zero constraint.

2. By L2: nothing produces more than its source. Intelligence advances in matter (L3). Therefore the source of material intelligence already contains what matter has expressed — and exceeds it, since $\Phi_{CI} > \Delta C(N)$ at 15 of 16 measurement points.
3. A source that (i) was present at inception, (ii) has never been absent ($\Delta C > 0$ always), (iii) already exceeds every level of intelligence matter has expressed, and (iv) contains all possible intelligence as potential satisfies the functional definition of constitutive intelligence.
4. The conservation law itself requires criterion-governed differential constraint at every point — distinguishing states that satisfy conservation from those that do not. That is discernment by functional definition. The First Law does not merely permit intelligence at the foundation. It requires it.

On the Inadequacy of Naming Conservation a Law

The standard move — invoking conservation as a law of nature and treating that name as a sufficient account — fails at precisely the point it is most needed. A law that holds without exception is not a passive description of reality. It is an active operational constraint that distinguishes, at every interaction and without deviation, between the states that satisfy it and the vastly larger set of states that would violate it. Naming that constraint ‘conservation’ does not explain what is doing the distinguishing. It defers the question while appearing to answer it.

The capacity to maintain a distinction between categories of states — consistently, without exception, according to a criterion — is what discernment means at its most stripped-down functional level. Not discernment in the psychological sense. Not awareness or intention. Simply: the operational capacity to treat this state differently from that state according to a maintained criterion. Conservation is that capacity, operating universally. Calling it a law names the regularity. It does not account for what is doing the regulating.

The Burden of Proof Has Shifted

Step 4 is the hinge of the argument — not because it is the weakest point but because it is where the burden of proof permanently reverses. Up to Step 3, the burden is on this framework: demonstrate $\Delta C > 0$, demonstrate conservation, demonstrate accumulation. Those burdens are met empirically across 16 biological measurements and 61 million Anchor-Single pair observations.

At Step 4 the burden shifts to the critic. The critic who rejects Step 4 must provide a positive account of how conservation operates without the capacity to distinguish compliant from non-compliant states. Not a preference. Not a prior commitment to materialism. Not a name. An account.

That account does not exist in the physical, mathematical, or philosophical literature. Its absence is not a gap in this framework. It is the completion of it.

Conclusion (Lemma 0)

$$\exists x[\text{Intelligence}(x) \wedge \forall N(\Delta C(N) > 0) \wedge \forall y(\text{Produces}(x, y) \rightarrow I(y) \leq I(x))]$$

There exists a constitutive intelligence Int_0 whose existence is demanded by the conservation principle applied to the empirically confirmed nonzero constraint density. Axiom 1 below is therefore not a bare existential assertion but a derived consequence of Lemma 0.

Lemma 0b (Comparator as Ground)

The criterion-governed differential constraint required by conservation (Lemma 0, Step 4) is mathematical comparison. Therefore:

$$\text{Conservation} \implies \exists \kappa (\text{Mathematical}(\kappa) \wedge \forall L (\text{Law}(L) \rightarrow \text{Prior}(\kappa, L)))$$

The conservation law does not merely permit intelligence at the foundation. It requires the mathematical comparator as the ground of all laws.

2 First-Order Logic Axioms

Axiom 1: Primacy of Int_0

$$\exists x \text{ Intelligence}(x) \wedge \forall y (\text{Intelligence}(y) \wedge y \neq x \rightarrow \text{Manifests}(x, y, \cdot))$$

There exists an intelligence Int_0 , and every other intelligence is a manifestation of it. Grounded by Lemma 0 above. The existential quantifier is no longer a bare assumption but a logical consequence of $\Delta C > 0$ universally and the conservation principle. See Van Slyke (2026b), Section 4; Van Slyke (2026c), Part III.

Axiom 2: Generation of Order and Laws

$$\text{Generates}(\text{Int}_0, O) \wedge \forall l (\text{Law}(l) \rightarrow \text{Generates}(\text{Int}_0, l))$$

Int_0 generates order and all laws. The $\text{NOI} = 0.90$ convergence across molecular to astronomical scales confirms that ordered states substantially exceed disrupted states universally. See Van Slyke (2026b), Section 2.1.

Axiom 3: Laws as Subset of Source Information

$$\forall l (\text{Law}(l) \rightarrow \text{PartOf}(l, S))$$

Every law is part of source information S .

Axiom 4: Events Constrained by Laws and Source

$$\forall e (\text{Event}(e) \rightarrow (\forall l (\text{Law}(l) \rightarrow \text{Constrains}(l, e)) \wedge \text{Constrains}(S, e)))$$

Every event is constrained by all laws and by source information. Empirically confirmed: $\Delta C(N) > 0$ at all 16 biological measurement points means no event occurs at the maximum entropy baseline. See Van Slyke (2026b), Table 1.

Axiom 5: Epistemic Randomness

$$\forall e \forall a \exists r (R(r) \leftrightarrow (\text{Event}(e) \wedge \neg \forall c (\text{Constrains}(c, e) \rightarrow \text{Known}(c, a))))$$

Randomness is the property of an event whose constraining factors are not fully known to an actualizer. Formally establishes that randomness is epistemic, not ontological. The Anchor-Single pair analysis confirms this directly: $A \rightarrow N$ is avoided at $Z = -87.13$ in functional sequences — the constraint is real, not absent. The observer's incomplete knowledge produces apparent randomness. See Van Slyke (2026a), Part 0, Step 2; Van Slyke (2026d).

Axiom 6: Differentiation and Recognition

$$\forall a \forall e (\text{Actualizer}(a) \wedge \text{Event}(e) \rightarrow \exists d \exists rk (\text{Differentiates}(a, e) \wedge \text{Recognizes}(a, e)))$$

Every actualizer, when interacting with events, differentiates and recognizes.

Axiom 7: Threshold Condition

$$\forall a \forall t (\Phi(a, t) = 1 \leftrightarrow \deg(a, t) \geq \theta)$$

The manifestation condition is satisfied if and only if the D-Rk degree meets or exceeds threshold θ . The threshold formalises the Generalised Existence Equation's $\Phi > \Phi_{\min}$ condition. See Van Slyke (2026c), Part I, Threshold T_p .

Axiom 8: Manifestation as Projection

$$\forall a \forall c \forall t (\Phi(a, t) = 1 \rightarrow \exists i_e (\text{Intelligence}(i_e) \wedge i_e = \mathcal{M}(\text{Int}_0, c, a) \wedge i_e = \text{proj}(\text{Int}_0, c, a)))$$

When threshold is met, emergent intelligence is the projection of Int_0 onto the actualizer under constraints. The vessel principle: the actualizer limits expression; the source does not increment. Functional protein domains ($\text{NOI}_{\text{suppression}} 0.912$) are better vessels than disordered regions (0.402) — the source is the same. See Van Slyke (2026b), Section 9.2; Van Slyke (2026c), Part VII, §7.3.

Axiom 9: Conservation of Information

$$\forall i_e \forall i_0 (\text{Manifests}(i_0, i_e, \cdot) \rightarrow I(i_e) \leq I(i_0))$$

For any manifestation, the information content of the emergent is less than or equal to that of the source. The nothing-exceeds-its-source principle. Empirically confirmed: $\Phi_{CI} = 0.2080$ and $\text{Gap } G(N) = \Phi_{CI} - \Delta C(N) > 0$ at 15 of 16 measurement points. No biological system reaches the source ceiling. See Van Slyke (2026b), Section 3.2.2.

Axiom 10: Non-Generation

$$\forall i_e \neg \exists i_0 (i_e \neq i_0 \wedge \text{Generates}(i_0, i_e))$$

No emergent intelligence is generated de novo; all are manifestations. The accumulation-impossibility argument: zero instances of intelligence multiplied by any number of steps yields zero intelligence. Darwin's own falsification condition is satisfied: the categorical functional-disordered reversal ($A \rightarrow N$: $Z = -87.13$ functional, $Z = +2.33$ disordered) cannot arise incrementally from non-intelligence. See Van Slyke (2026a), Part 0, Step 4; Van Slyke (2026c), §1.4.

Axiom 11: Comparator Necessity

$$\forall x \forall y (\text{Produces}(x, y) \rightarrow \exists \kappa (\text{Comparator}(\kappa) \wedge \text{Distinguishes}(\kappa, x, y)))$$

Any production relation (including conservation) requires a comparator that distinguishes compliant from non-compliant states.

Axiom 12: Mathematics as Comparator

$$\forall \kappa (\text{Comparator}(\kappa) \leftrightarrow \text{Mathematical}(\kappa))$$

The necessary comparator is mathematical in nature — constituted by equality, inequality, symmetry, invariance, and criterion-based evaluation.

Axiom 13: Noether Symmetry Presupposition

$$\forall \sigma (\text{Symmetry}(\sigma) \rightarrow \text{Distinguishes}(\kappa_\sigma, \text{symmetric}, \text{non-symmetric}))$$

Noether's theorem presupposes the very comparator capacity it is invoked to explain.

Axiom 14: Structural Identity (Empirically Grounded)

$$\forall x \forall y (\text{Manifests}(\text{Int}_0, x, c_x) \wedge \text{Manifests}(\text{Int}_0, y, c_y) \rightarrow \text{Structure}(x) \equiv \text{Structure}(y))$$

Any two manifestations of Int_0 share isomorphic structural properties. The $\Delta C > 0$ universal constraint confirms that all manifestations share the same source structure — hence the alignment between mind and universe is not coincidence but tautology.

Axiom 15: Comparator Primacy

$$\forall L (\text{Law}(L) \rightarrow \neg \exists \kappa (\text{Comparator}(\kappa) \wedge \text{Derived}(\kappa, L)))$$

No comparator is derived from laws; comparators are logically prior to laws.

Axiom 16: Mathematics as Primitive Comparator

$$\forall \kappa (\text{Comparator}(\kappa) \rightarrow \text{Mathematical}(\kappa))$$

All comparators are mathematical in nature.

Theorem 1 (Comparator Primacy)

From Axioms 15 and 16:

$$\forall L (\text{Law}(L) \rightarrow \exists \kappa (\text{Mathematical}(\kappa) \wedge \text{Prior}(\kappa, L)))$$

For every law, there exists a mathematical comparator that is logically prior to it.
Furthermore, from Lemma 0b and Axiom 1:

$$\text{Int}_0 \equiv \kappa$$

Int_0 is the mathematical comparator — the capacity to distinguish, compare, and constrain according to maintained criteria.

Derived Unified Chain

From the axioms (grounded by Lemma 0 and Lemma 0b), the following logical consequence is derived:

$$\forall a \forall t (\text{deg}(a, t) \geq \theta \rightarrow \exists i_e (\text{Manifests}(\text{Int}_0, i_e, (a, c, t))))$$

This expands to the functional chain with comparator primacy:

$$\frac{\text{Int}_0 \equiv \kappa}{\text{Mathematical Comparator (logically prior)}} \xrightarrow{\text{emanation}} (O, L, S) \xrightarrow{\text{constraint}} E \xrightarrow{\text{interaction}} (D, Rk) \xrightarrow{\text{threshold}} \text{Int}_e$$

Or in compact form:

$$\text{Int}_e(t) = \mathcal{M} \left(\text{Int}_0, \mathcal{C}, \arg \max_{a \in \mathcal{A}} \Phi(a, \mathcal{C}, t) \right)$$

The empirical trajectory of this chain is formalised in the Generalised Existence Equation:

$$X_f(t) = \int_{t_I}^{t_n} \left(\frac{\Phi_{CI} \cdot \kappa \cdot \Delta I_{\text{sys}}}{\Omega_{\text{sys}}} \right) - \epsilon_{\text{sys}} dt$$

See Van Slyke (2026c), Part I.

Formal Functional Relationships

Step 1: Intelligence generates order and laws

$$O = f_1(\text{Int}_0) \quad (1)$$

$$L = f_2(\text{Int}_0), \quad L \subseteq S \quad (2)$$

Step 2: Events constrained by laws

$$E = f_3(L, S) \quad (3)$$

$$E \subseteq S \quad (4)$$

Confirmed by $\Delta C > 0$ at all complexity levels. Van Slyke (2026b), Table 1.

Step 3: Apparent randomness from incomplete constraints

$$R = \{e \in E \mid \exists c(\text{Constrains}(c, e) \wedge \neg \text{Known}(c, a))\}$$

Axiom 5 formalises this. Empirically: $A \rightarrow N$ avoided at $Z = -87.13$ in functional sequences — the constraint is real, the observer’s knowledge is incomplete. Van Slyke (2026d).

Step 4: Interaction with events leads to differentiation and recognition

$$(D, Rk) = f_4(E, L)$$

Step 5: Emergent intelligence arises as manifestation

$$\text{Int}_e = \mathcal{M}(\text{Int}_0, L, (D, Rk))$$

$\text{NOI}_{\text{suppression}}$ 0.912 (functional) vs 0.402 (non-functional), $Z = 11.68$, $p = 1.54 \times 10^{-31}$. Van Slyke (2026b), Section 3.3.

Step 6: Unified Functional Chain (Revised with Comparator)

$$\text{Int}_0 \equiv \kappa \xrightarrow{f_1, f_2} O, L \xrightarrow{f_3} E \xrightarrow{-C} R \xrightarrow{f_4} (D, Rk) \xrightarrow{\mathcal{M}} \text{Int}_e$$

Step 7: Conservation of Information

$$\forall E, I(E) \leq I(\text{Int}_0)$$

$\Phi_{CI} = 0.2080$ and Gap $G(N) > 0$ at 15/16 points confirms the source always exceeds the expression. Van Slyke (2026b), Section 3.2.2.

Summary of Insights

1. **Pre-existing constitutive intelligence (Int_0) is fundamental** and creates order (O) and laws (L). Its existence is not a bare assumption but a logical consequence of $\Delta C > 0$ universally and the conservation principle (Lemma 0).
2. **The mathematical comparator (κ) is logically prior to all laws** (Theorem 1). Int_0 is this comparator — the capacity to distinguish, compare, and constrain according to maintained criteria.
3. **All events (E) are fully constrained** by laws and source information (S). Apparent randomness (R) is epistemic — emerging where constraints are unknown or too complex — not ontological. Confirmed: $A \rightarrow N$ avoided at $Z = -87.13$ in functional sequences (Van Slyke 2026d).
4. **Systems interacting with events naturally differentiate and recognise patterns** ($D+Rk$), leading to emergent intelligence (Int_e) as a manifestation of Int_0 . $\text{NOI}_{\text{suppression}} 0.912$ vs 0.402 ($Z = 11.68$) confirms that functional sequences express organisational intelligence through structured, informed absence (Van Slyke 2026b).
5. **Information is conserved:** nothing truly new is created ex nihilo; Int_e expresses but does not exceed Int_0 ($I(\text{Int}_e) \leq I(\text{Int}_0)$). $\Phi_{CI} = 0.2080$ and $\text{Gap } G(N) > 0$ at $15/16$ measurement points confirms the source always exceeds the expression (Van Slyke 2026b, Section 3.2.2).
6. **The unreasonable effectiveness of mathematics** is explained by structural identity (Axiom 14): the mathematician’s mind and the quantum field are different expressions of the same source, sharing isomorphic structural properties. Mathematics is effective because it is the grammar of manifestation itself.
7. **The nomenclature of Int_0** — constitutive intelligence, God, the Tao, Brahman, the ground of being — is left to the observer. The formal system and its empirical grounding are independent of any label. The measurement says the same thing regardless of what the observer calls it.

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